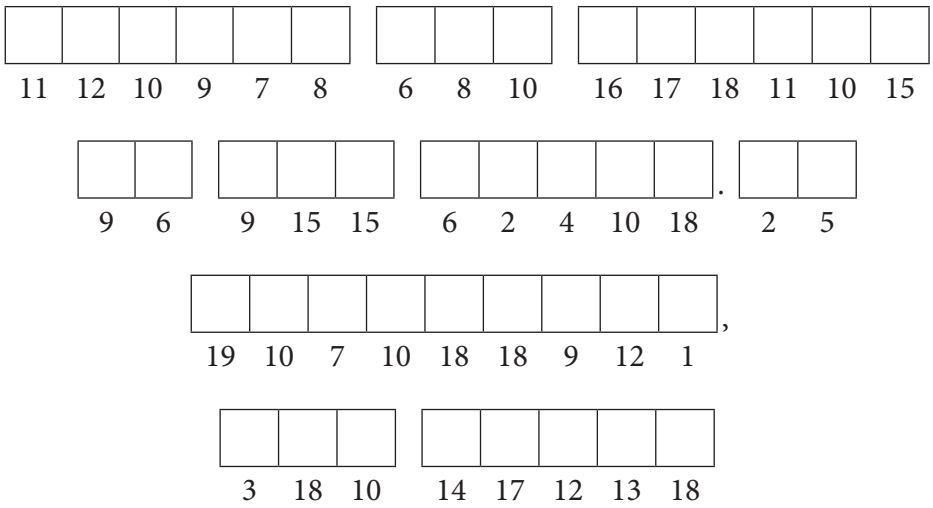
Name: _____

WORKSHEET

Classifying sequences



St Francis of Assisi

Rules

An **arithmetic progression** is a sequence where consecutive terms differ by the same amount.
Example: 2, 4, 6, 8, … Each term differs by +2.
A **geometric progression** is a sequence where the ratio of successive terms to each other is the same.
Example: 2, 4, 8, 16, … Each term has a common ratio of 2 to the term before.

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Questions

Classify each of the following sequences as an arithmetic progression, geometric progression or neither.

1 7, 14, 28, 56, …

2 3, 6, 9, 12, …

3 $-1, 2, -4, 8, \ldots$

4 5, 8, 11, 15, …

5 $-10, 5, -2\frac{1}{2}, 1\frac{1}{4}, \ldots$

6 1, 5, 9, 13, …

7 7, 4, 1, −2, …

8 6, −1, −8, −15, …

9 81, 27, 9, 3, …

10 4, 7, 9, 12, …

11 $-2, 1\frac{1}{2}, -\frac{9}{16}, \frac{13}{25}, \ldots$

12 8, −12, 18, −27, …

13 $1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \ldots$

14 $-2, \frac{1}{2}, 3, 5\frac{1}{2}, \ldots$

15 $\frac{1}{5}, 1, 5, 25, \ldots$

16 −27, 9, −3, 1, …

17 $3, 1\frac{3}{4}, \frac{1}{2}, -\frac{3}{4}, \ldots$

18 4, −6, 9, −15, …

19 −5, −9, −13, −17, …

Solutions

Question	Arithmetic	Geometric	Neither
1	E	Y	C
2	I	N	K
3	A	U	F
4	P	B	M
5	L	F	S
6	T	O	V
7	C	D	I
8	H	J	W
9	N	A	G
10	O	K	E
11	Q	X	P
12	F	R	T
13	S	V	D
14	W	I	B
15	Z	L	K
16	M	G	U
17	O	T	J
18	H	A	S
19	N	Y	V

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